

Remote work, team velocity, and long-term capability

An agent-based simulation of a software-development team

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Abstract

This report presents an agent-based model (ABM) of a software-development team built to study how the share of remote work affects team output and long-term capability. Each developer is an agent with a per-tick state machine; blockers are resolved either alone or with a colleague's help, and help between two people in the office is fast and synchronous while any help involving someone at home is slow and asynchronous. The model is examined at two horizons. Over a *single sprint*, with skill held fixed, velocity falls smoothly with the remote share, and the size of the penalty depends strongly on how often help is actually needed: -3.6% for a high-skill team up to -13.2% for a team that blocks frequently. Over a *year*, with learning and turnover added, a co-located team delivers only about 2.5–2.8% more cumulative output than a fully remote one, and, against the naive prior, this penalty is *front-loaded*: it is largest while the team is junior and shrinks as the team climbs the learning curve. A single-parameter sensitivity analysis confirms that the average wait for help rises by 59% across the remote range while throughput falls by only 2.5%, and that the most decision-relevant effect of remote work is on the *spread* of yearly output (roughly tripling), not on its mean. The central conclusion is that the cost of remote work is not a fixed number but is contingent on the state of the team.

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1 Introduction

Whether a software team should work in the office, fully remotely, or somewhere in between is a recurring management decision, and the evidence behind it is mixed: different studies report remote work as neutral, mildly harmful, or even helpful for output, and as both good and bad for retention. A decision-maker therefore needs more than a single average effect; they need to know *under which conditions* remote work is cheap and under which it is costly.

This report builds an agent-based model to answer that question for one team. The reason for choosing ABM rather than a regression on aggregate data is that the quantity of interest, team throughput, emerges from many small, local interactions: an individual developer hits a blocker, tries to clear it alone, and only then reaches for a colleague, whose availability and location decide whether the help is fast or slow. These micro-events accumulate into the macroscopic velocity curve. ABM lets us write down the local rules, which are relatively well understood, and *observe* the aggregate consequence, which is not. As we will see, the model produces at least one result that reverses the intuitive prior, which is exactly the kind of non-obvious, emergent finding that justifies the method.

The single policy variable under study throughout is `remote_share`: the probability that each developer works from home on a given day. Everything else is held at a defensible base case, and the report asks how output and capability respond as `remote_share` is swept from a fully co-located team (0) to a fully remote one (1).

The report is organised as follows. Section 2 describes the model and its per-tick state machine, and grounds its parameters in the empirical literature. Section 3 measures the direct effect of remote work over a single sprint with skill fixed. Section 4 extends the horizon to a year and adds learning and turnover. Section 5 reports a formal one-parameter sensitivity analysis. Section 6 describes the interactive interface, and Section 8 draws the conclusions together.

2 The model

2.1 Overview

The simulation is built on Mesa 3.x. Time advances in discrete *ticks* of 15 minutes; a working day is 32 ticks (eight hours) and a sprint of ten working days is 320 ticks. A team of developers works through a backlog of tasks whose sizes are story points drawn from $\{1, 2, 3, 5\}$, each point costing four ticks of work by default. Tasks are passive data; the developers are the agents. The headline experiments use a team of ten developers.

Each morning every developer is independently assigned to the office or to home, with probability `remote_share` of working from home. This single daily coin-flip is the only thing that `remote_share` controls directly; its entire effect on output flows from how that location choice changes the speed (and, later, the educational value) of help between colleagues.

2.2 The developer state machine

The core of the model is a per-tick state machine run by every developer. Its top-level state is one of *idle*, *working*, *blocked*, or *helping*; while *blocked*, a secondary stage (*solo*, *ask_next_tick*, *waiting_for_help*) tracks how the developer is trying to get unstuck. Figure 1 shows the full cycle.

The logic is as follows. An *idle* developer pulls the next task from the backlog and starts *working*. A *working* developer reduces the task’s remaining work by one tick; if the task is finished it is counted towards velocity and the developer returns to *idle*, otherwise the developer may hit a blocker with probability `block_prob` on each tick. A *blocked* developer first attempts a solo resolution, succeeding with probability equal to their individual *solo skill*. On failure they look, on the next tick, for an available colleague (one who is working or *idle*). If a developer in

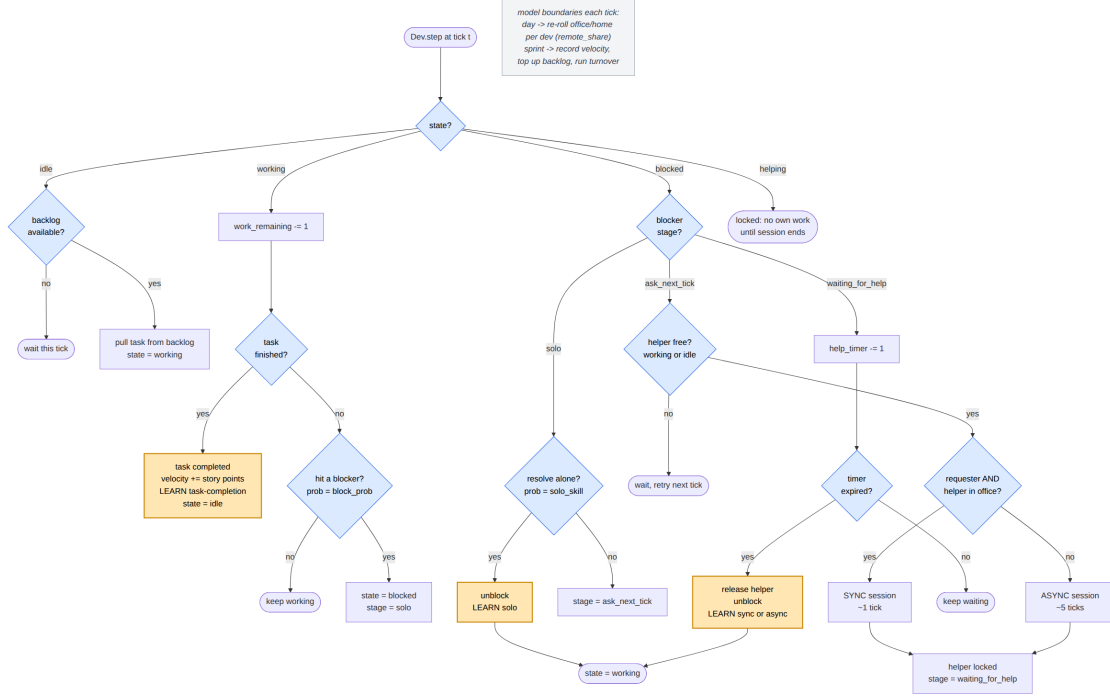


Figure 1: The per-tick developer state machine. An idle developer pulls work; a working developer makes progress and may hit a blocker; a blocked developer first tries to resolve it alone, then recruits a helper, opening a fast synchronous session if both are in the office or a slow asynchronous one otherwise. Shaded boxes mark *learning events*: the three moments at which a developer’s skill grows. The note lists the model-level boundaries that wrap the per-agent loop.

the office finds an office colleague the help is *synchronous* and short (1 ± 1 ticks); if either party is at home it is *asynchronous* and long (5 ± 2 ticks). The helper is locked for the duration of the session and cannot do their own work, which is the channel through which one person’s blocker reduces the whole team’s throughput. When no helper is free the developer simply waits and retries; this is a deliberate throughput limit, not a modelling oversight.

This single distinction, that synchronous, co-located help is roughly five times faster than asynchronous, remote help, is the entire mechanism by which remote work affects a sprint’s output. Everything in Part A follows from it.

2.3 Learning

Over a single sprint it is reasonable to treat each developer’s solo skill as fixed, but over a year it is not, because people get better at their work. In the extended model, solo skill is the probability of clearing a blocker alone, and it grows with experience. Every time a developer completes a task or resolves a blocker, their skill takes a small, deterministic step towards a ceiling `skill_cap`, and the steps shrink as the ceiling approaches:

$$\text{skill} \leftarrow \text{skill} + \text{learning_rate} \cdot w \cdot (\text{skill_cap} - \text{skill}),$$

where w is a weight that depends on *how* the experience was gained. This diminishing-returns form produces fast early progress and then a plateau, matching the empirical onboarding pattern in which a new hire reaches most of their productivity within roughly six months.

The weight w is where remote work re-enters. Resolving a blocker through co-located help teaches more than resolving the same blocker asynchronously (`sync_help_weight` = 1.5 versus `async_help_weight` = 1.0). This is the only route by which remote work slows learning, so the model needs no separate “remote learning” switch. Crucially, most learning weight comes

from ordinary task completion, which is location-independent; the location-dependent mentoring bonus applies only to the rarer blocker-resolution events.

2.4 Turnover

At the end of each sprint every developer may leave, with a per-sprint hazard derived from an annual attrition rate (about 12% per year in the base case). Anyone who leaves is replaced immediately by a junior who starts at low skill and must learn from scratch; any partly finished task they held returns to the backlog with its progress lost. Each departure therefore pulls the team’s mean skill down and resets part of the learning curve.

Remote work can raise or lower the chance of leaving through a single coefficient, `remote_attrition_coef`. Because the empirical sign is genuinely contested (one large randomized trial finds remote work *improves* retention, while a natural experiment finds lost in-person mentorship *raises* junior quitting), the coefficient defaults to zero and results are reported under both readings. This contested knob is the headline decision-support lever of the model.

2.5 Parameters and their empirical grounding

The model’s parameters are not guessed; each is a defensible synthesis of published evidence, intended as a sensitivity-analysis base case rather than a precise constant. Table 1 lists the load-bearing ones with their values, ranges, and the strength of the evidence behind them. No source measures the model’s exact constructs (population-level voluntary developer turnover, or the probability of resolving a blocker solo as a function of tenure), so every value is a reconstruction from adjacent measures, and the report relies on sweeps rather than on any single point estimate.

Table 1: The load-bearing parameters, their base-case values, and their empirical grounding.

Parameter	Base case	Sweep / range	Grounding (evidence strength)
<code>remote_share</code>	0.5	0–1	Policy variable under study.
<code>block_prob</code>	0.02 / tick	0.01–0.05	Process rate; swept as a moderator.
<code>mean_solo_skill</code>	0.5	0.3–0.7	Team-mean blocker self-resolution.
<code>sync_help_mean</code>	1 tick	fixed	Co-located help is fast.
<code>async_help_mean</code>	5 ticks	fixed	Remote help $\sim 5\times$ slower (async).
<code>annual_attrition</code>	0.12	0.08–0.20	Med: $\sim 10\text{--}15\%$ /yr developer voluntary turnover; no clean source.
<code>skill_cap</code>	0.90	0.85–0.95	High: early hard plateau (3 independent sources).
<code>junior_mean_skill</code>	$0.40 \times \text{cap}$	0.35–0.50	Med: reconstructed from expert/novice $\sim 2:1$.
<code>learning_rate</code>	0.0123	calibrated	Med: fitted so a junior closes $\sim 90\%$ of the gap in ~ 6 months.
<code>sync : async_weight</code>	1.5 : 1.0	$1.2 \times \text{--} 2 \times$	High (general), Low (junior multiplier): co-located mentoring premium.
<code>remote_attrition_coef</code>	0.0	$-1 \dots +1$	High both arms, sign contested (retention vs. isolation).

The learning rate deserves a note because it is the one parameter fitted by the model itself rather than read off the literature. It was calibrated by simulation so that a junior closes about 90% of the gap to the skill ceiling in roughly twelve sprints (~ 6 months), the modal ramp-up time in onboarding studies; the bisection result `learning_rate` = 0.0123 agrees with an independent analytic cross-check. The fitted rate reproduces the documented ramp: 68% of the gap closed by three months, 90% by six, 99% by twelve.

3 Part A: a single sprint

Part A holds each developer’s skill fixed and measures the direct effect of remote work on one sprint’s velocity. Three experiment series were run, each with 20 replications (different random seeds) per cell, for 940 simulations in total. The remaining parameters keep their defaults: office help takes 1 ± 1 ticks, remote help 5 ± 2 ticks, and individual solo skill is spread by ± 0.2 around its mean.

#	Factor(s)	Levels	Replications	Total runs
1	<code>remote_share</code> (baseline)	11 (0.0–1.0, step 0.1)	20	220
2	<code>remote_share</code> $\times$ <code>mean_solo_skill</code>	6×3 (skill 0.3, 0.5, 0.7)	20	360
3	<code>remote_share</code> $\times$ <code>block_prob</code>	6×3 (bp 0.01, 0.02, 0.05)	20	360

3.1 Experiment 1: baseline sweep

As the share of remote work rises from none to everyone, velocity falls steadily and the average wait for help grows, because more of the help that does happen is the slower, asynchronous kind (Figure 2). The fall is smooth: there is no threshold or sudden collapse.

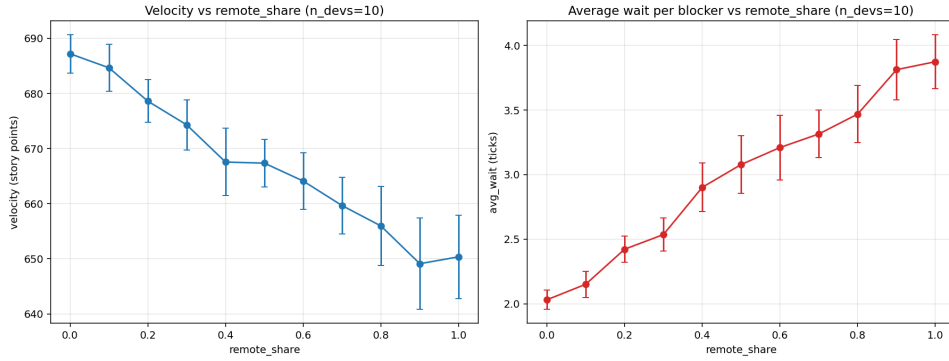


Figure 2: Velocity and average wait per blocker against the share of remote work, baseline sweep (10 developers, 20 replications per level, 95% confidence intervals).

<code>remote_share</code>	velocity (mean $\pm$ CI95)	avg. wait	blockers resolved
0.0	687.2 $\pm$ 3.5	2.0	52.0
0.3	674.3 $\pm$ 4.6	2.5	52.3
0.5	667.4 $\pm$ 4.3	3.1	49.2
0.7	659.6 $\pm$ 5.2	3.3	50.6
1.0	650.4 $\pm$ 7.6	3.9	47.5

3.2 Experiment 2: remote share and individual skill

The size of the remote penalty depends on how often people actually need help. A more skilled team clears most blockers alone and barely notices slower remote help; a less skilled team leans on the help channel and suffers far more. The velocity lost when moving from a fully office to a fully remote team ranges from 9.0% for a low-skill team down to 3.6% for a high-skill one (Figure 3).

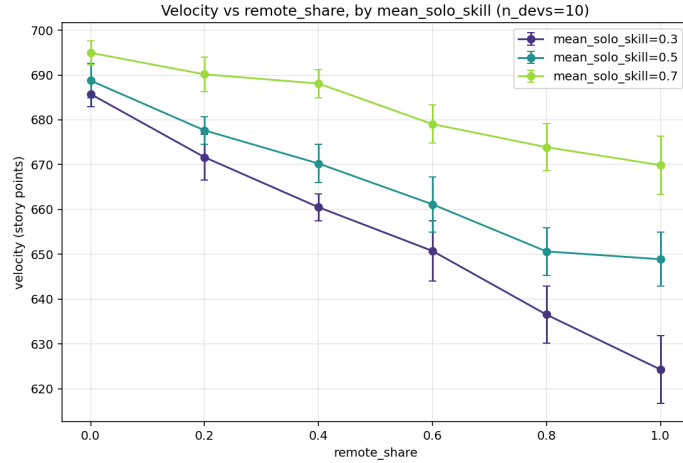


Figure 3: Velocity against remote share, grouped by mean solo skill.

mean_solo_skill	velocity @ 0 %	velocity @ 100 %	absolute loss	relative loss
0.30	685.8	624.3	−61.5 SP	−9.0%
0.50	688.8	648.9	−39.9 SP	−5.8%
0.70	695.0	669.9	−25.1 SP	−3.6%

3.3 Experiment 3: remote share and blocker frequency

For the same reason, the penalty grows sharply with how often blockers occur. When blockers are rare the help channel is used little and remote work costs about 3%; when they are frequent the team is constantly waiting for help and the cost rises to roughly 13% (Figure 4). Experiments 2 and 3 tell one story from two directions: the remote penalty is a tax on the help channel, so it scales with how heavily that channel is used.

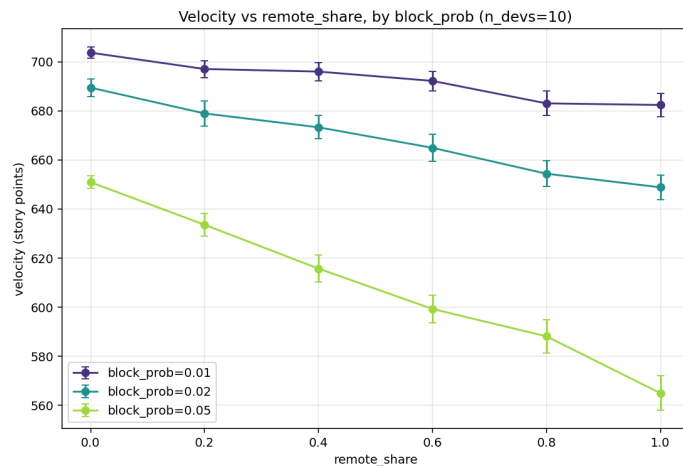


Figure 4: Velocity against remote share, grouped by blocker probability.

block_prob	velocity @ 0 %	velocity @ 100 %	absolute loss	relative loss
0.01	703.7	682.4	−21.3 SP	−3.0%
0.02	689.5	648.9	−40.6 SP	−5.9%
0.05	651.0	565.0	−86.0 SP	−13.2%

4 Part B: a year, with learning and turnover

Part A treats skill as fixed, which is fine for one sprint but not for a year, over which people get better and teams lose and gain members. Part B turns on learning and turnover and runs the model over 24 two-week sprints (~ 1 year). The measure that matters here is *cumulative velocity*: the total story points delivered over the whole year, not in any single sprint. Four sweeps of the remote share were run, again with 20 replications per cell.

4.1 The remote penalty is front-loaded, not compounding

Across the year a fully co-located team delivers roughly 2.5–2.8% more than a fully remote one. The gap grows monotonically with the remote share, but its most telling feature is *when* it occurs: it is largest early in the year and shrinks as the team matures.

Horizon	Advantage of the office team
6 sprints (about 3 months)	+4.7%
12 sprints (about 6 months)	+3.3%
24 sprints (about 1 year)	+2.6%

It was natural to expect the opposite: that a small per-sprint penalty would *compound* as the two teams’ skills diverged over the year. The model shows the reverse, and the reason is instructive. The remote penalty is at heart a penalty on the help channel: remote developers wait longer for asynchronous help. Learning works *against* this penalty rather than with it. As both teams grow more skilled they need help less often, so the channel where remote work loses ground matters less and less. With learning switched off, the office team’s advantage over a year is 6.2%; with learning on it falls to 2.5% (Figure 5). The cost of remote work is concentrated in the early, junior, frequently-blocked phase and fades once the team is experienced.

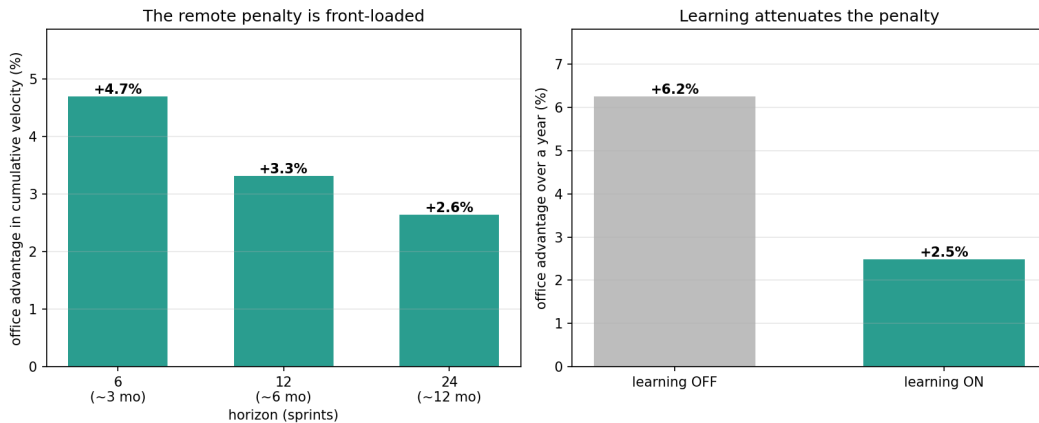


Figure 5: The office advantage in cumulative velocity is largest early and shrinks over the year (left); switching learning on more than halves it (right).

The skill ramp makes the mechanism visible (Figure 6). For a team that starts as juniors, the office team ramps a little faster, but both reach the same ceiling, so the office/remote skill gap is widest mid-ramp and closes by year end. This is why the penalty shows up in ramp speed and cumulative output, not in the end-of-year skill level.

4.2 Turnover reproduces both literature scenarios

The turnover coefficient behaves as designed. Under the *retention* reading, where remote work helps keep people, a fully remote team loses fewer members over the year than an office team;

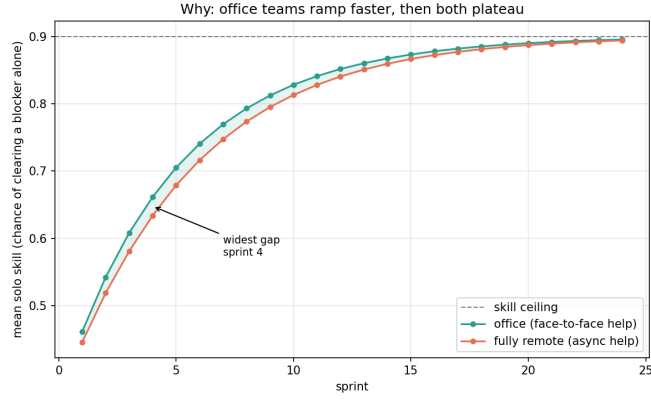


Figure 6: Mean solo skill over the year for a team of juniors. The office team ramps slightly faster, but both converge to the same plateau.

under the *isolation* reading, where remote work pushes juniors out, it loses more (Figure 7). Reporting both arms, rather than picking one, is the core decision-support output of the model, because the empirical sign is genuinely unresolved.

Scenario	departures: office $\rightarrow$ fully remote
retention (remote keeps people)	1.9 $\rightarrow$ 0.7
isolation (remote pushes them out)	0.6 $\rightarrow$ 2.0

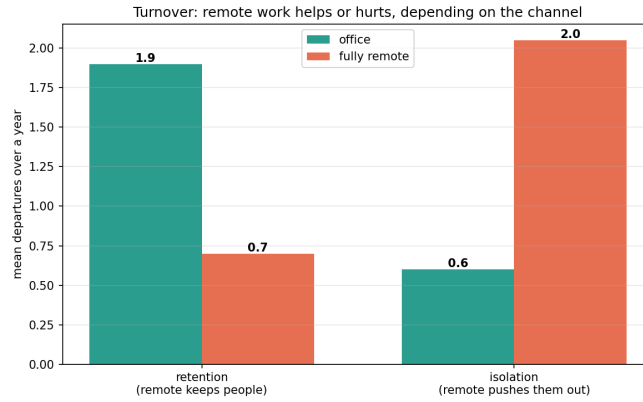


Figure 7: Departures over a year, office versus fully remote, under the two readings of how remote work affects retention.

Under immediate junior replacement, turnover is nearly velocity-neutral within a single year, so its signal lives in the departure count rather than in throughput. Over multiple years, however, a departure that swaps a senior for a junior would feed back into the skill channel, so even a moderate effect through `remote_attrition_coef` would plausibly outweigh the 2.5% throughput effect studied here.

5 Sensitivity analysis of `remote_share`

To put the headline parameter on a firmer footing, a formal one-parameter sensitivity study was run in the year-long regime (24 sprints, learning and turnover on, all other parameters at their defaults). `remote_share` was swept across its full range $[0, 1]$ in steps of 0.05, with 80 repetitions

per level; a separate sub-study with 400 repetitions at three levels was used to check how many repetitions are needed.

5.1 How many repetitions are enough

The number of repetitions needed is not the same for every quantity (Figure 8). A full year averages over many thousands of small events, so the mean of yearly throughput settles quickly: about 40 repetitions give it to within $\pm 0.1\%$, and 80 give $\pm 0.08\%$ even at the noisiest end of the range. A count of rare events behaves very differently. The attrition count has a mean near 1.1 and a standard deviation near 1.0, so its relative error is still about $\pm 20\%$ at 80 repetitions. Eighty repetitions are therefore ample for the throughput results, but a rare-event count would need several hundred.

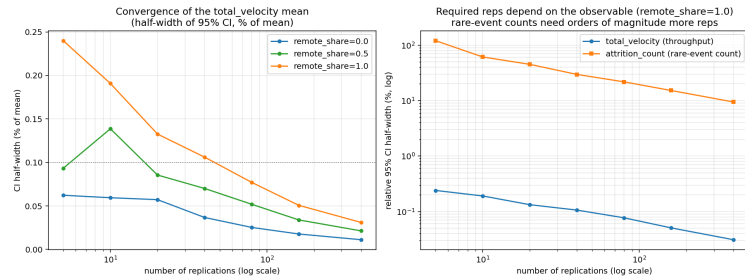


Figure 8: Width of the 95% confidence interval against the number of repetitions. Throughput (left) settles quickly; the attrition count (right) needs far more runs for the same relative precision.

5.2 The mean moves little; the spread moves a lot

Throughput goes down as remote work increases, but only by about 2.5% across the full range, with no sudden jump. The mechanism behind it changes much more: the average wait per blocker rises by 59% (Figure 9), because help involving someone at home is asynchronous and roughly five times slower. The two are loosely linked because blockers are rare next to ordinary work: the extra waiting amounts to about a thousand idle ticks against roughly 77,000 working ticks in the year, close to 1.3%, the same order of magnitude as the throughput drop. The parameter that looks most important in the mechanism is therefore not the one that controls the final result.

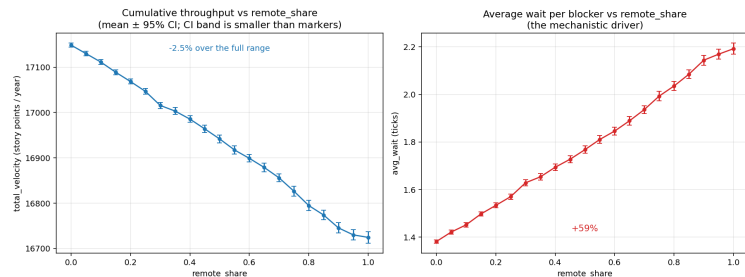


Figure 9: Yearly throughput falls by 2.5% across the full range (left), while the average wait per blocker rises by 59% (right). Means with 95% confidence intervals, 80 repetitions per point.

The mean alone, however, hides the most decision-relevant effect. Keeping the full set of run results rather than reducing each level to an average (Figure 10) shows that the spread of yearly throughput grows about threefold as remote work increases (the run-to-run standard deviation

risers from about 20 to 59 story points), while the mean moves by only 2.5%. In other words, how *predictable* the output is depends on remote work far more strongly than the average output does. An analysis that looked only at the mean would miss this entirely.

Two further results are worth stating. First, remote work does not reduce end-of-year skill, which stays flat near 0.88 across the whole range, because most learning comes from finishing tasks (location-independent) and the in-office mentoring bonus applies only to the rarer blocker-resolution events. Second, the attrition count shows no trend with `remote_share`, but only because the default `remote_attrition_coef`= 0 severs the link by construction. That flat result is an honest caveat, not a finding, and sweeping the coefficient over $[-1, 1]$ is the natural next step.

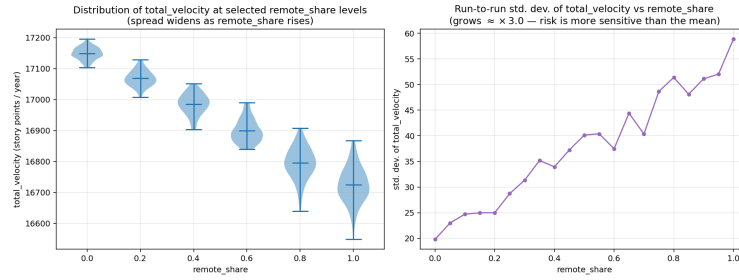


Figure 10: Spread of yearly throughput at each level (left) and its run-to-run standard deviation (right). The mean barely changes, but the spread grows about threefold.

6 The interactive interface

The model is driven through an interactive Streamlit application (`app_streamlit.py`, run with `streamlit run app_streamlit.py`). The page uses a wide layout split into a parameter *sidebar* on the left and a main area on the right that holds an animated view of the team followed by five metric charts. Changing any parameter re-runs the simulation and redraws everything. This section documents the whole interface: every parameter, the run-and-replay architecture, the animation, and each chart.

6.1 The parameter sidebar

The sidebar groups every model parameter into collapsible sections by role; the *Headline* group is expanded by default and the rest start collapsed. Each control is a slider with the default and range given in Table 2. The headline variable is `remote_share`; the remaining groups set up the team’s heterogeneity, the work process, learning, turnover, the run’s structure, and reproducibility.

Table 2: The complete sidebar: every parameter exposed by the Streamlit interface, with its default, range, and what it controls.

Parameter	Default (range)	What it controls
<i>Headline</i>		
<code>remote_share</code>	0.5 (0–1)	Probability each developer works from home on a given day, rolled afresh every morning. The policy variable under study.
<i>Team heterogeneity</i>		
<code>mean_solo_skill</code>	0.5 (0–1)	Team-mean solo skill: the chance a developer clears a blocker alone, without asking for help.

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Table 2 (continued)

Parameter	Default (range)	What it controls
solo_skill_spread	0.2 (0–0.5)	Half-width of the uniform spread of solo skill across developers, around the mean.
<i>Process</i>		
block_prob	0.02 (0–0.10)	Per-tick probability that a working developer hits a blocker.
sync_help_mean	1 (0–5)	Mean length, in ticks, of a co-located (office-to-office) help session.
async_help_mean	5 (1–20)	Mean length, in ticks, of a help session involving anyone working from home.
<i>Learning</i>		
learning_rate	0.0123 (0–0.05)	Size of the skill step taken on each learning event; set to 0 to switch learning off.
skill_cap	0.90 (0.5–1.0)	Ceiling that solo skill approaches but never exceeds.
sync_help_weight	1.5 (0–3)	Multiplier on the skill gained when a blocker is resolved through co-located help.
async_help_weight	1.0 (0–3)	Same multiplier for remote help; lower than the sync weight, so remote help teaches less.
<i>Turnover</i>		
annual_attrition	0.12 (0–0.30)	Annual probability that a developer leaves, converted internally to a per-sprint hazard.
junior_mean_skill	0.40 (0–0.8)	Skill of a newly hired replacement, as a fraction of the skill ceiling.
remote_attrition_coef	0.0 (–1–1)	Sign and size of remote work’s effect on leaving: positive raises attrition with remote share (isolation), negative lowers it (retention).
<i>Structural</i>		
n_devs	5 (2–10)	Number of developers on the team.
sprint_length	320 (32–640)	Ticks per sprint; 320 is ten working days at 32 ticks (15 minutes each) per day.
n_sprints	24 (1–52)	Number of sprints in the run; 24 two-week sprints is about a year.
sp_scaling_k	4 (1–16)	Ticks of work per story point.
<i>Reproducibility</i>		
seed	42 (0–10000)	Random-number seed; fixing it makes a run exactly repeatable.

Below these groups, separated by a divider, sits one control that belongs to the display rather than to the model: an *animation-frames* slider (100–1000, default 300) that caps how many ticks are captured for the animation. A long horizon is sub-sampled down to this many frames so that the playback stays responsive in the browser; finer sub-sampling gives smoother motion at a higher cost.

6.2 Run-once-then-replay architecture

Streamlit cannot animate a model while it is stepping, so the run and the playback are deliberately separated. On any change of parameters the model is executed once to completion and the state of every developer (activity, location, skill, and any active help session) is snapshotted at each captured tick. The resulting sequence of frames is then handed to Plotly, whose native play/pause buttons and time slider replay it entirely in the browser, with no recomputation while scrubbing. The whole run is cached on the parameter set (`st.cache_data`), so moving the slider or pressing play never re-runs the simulation; only changing a parameter does.

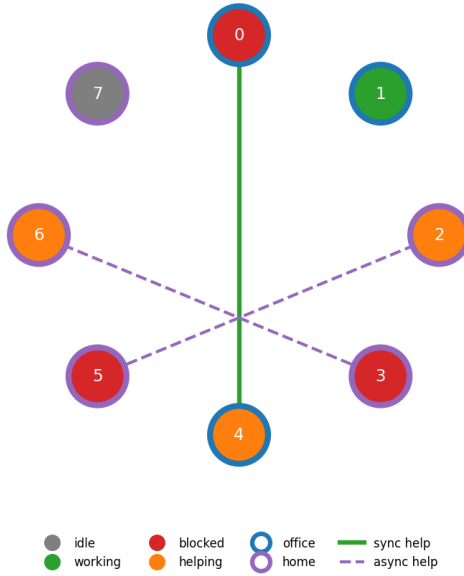


Figure 11: A single tick of the animated team view. Inner fill is the developer’s state and the outer ring is their location; a solid green edge is a co-located (sync) help session and a dashed purple edge a remote (async) one. Here a developer is being helped in the office while others, working from home, wait on slower asynchronous help.

6.3 The animated team view

Each developer is drawn as a node evenly spaced around a circle, labelled by its slot index (Figure 11). A node carries two independent colour channels. Its *inner fill* encodes what the developer is doing this tick: idle (grey), working (green), blocked (red), or helping someone else (orange). Its *outer ring* encodes where the developer is working today: in the office (blue) or from home (purple). Location is given its own visual channel because it is the single distinction that decides whether a help session is fast and co-located or slow and asynchronous. Hovering a node reveals its index, state, location, and current solo skill.

When one developer is blocked and another steps in to help, an *edge* is drawn between the two nodes for as long as the session lasts. The edge is a solid green line when both people are in the office (a synchronous session) and a dashed purple line when at least one is remote (an asynchronous one). A compact colour legend above the figure maps every fill, ring, and edge style. At a glance the picture therefore shows how much of the team is productive versus waiting, how the office/home split falls on any given day, and how much of the help that is happening is the slower remote kind: the very quantities whose averages drive the velocity and waiting-time curves below.

6.4 The metric charts

Below the animation, the main area shows five time-series charts read straight from the model’s data collector and laid out in two columns (Table 3). Together they connect the micro-level animation to the macro-level outcomes studied in the rest of the report: the velocity and waiting-time curves are the headline outputs, the developer-states curve is the aggregate of the node fills, and the skill and attrition curves expose the slower learning and turnover mechanisms.

Table 3: The five metric charts, all drawn from the data collector over ticks.

Chart	What it shows
Per-sprint velocity	Story points completed inside the current sprint, resetting to zero at each sprint boundary; the saw-tooth height is that sprint’s throughput.
Mean waiting time per resolved blocker	Running mean of the ticks between hitting a blocker and clearing it. It rises with the share of remote work, as more help arrives through the slower asynchronous channel.
Developer states over time	Number of developers in each state each tick (working, blocked, helping, idle), using the same colours as the node fills in the animation.
Team skill over time	Mean, minimum, and maximum solo skill across the team. The upward ramp is learning; the downward steps are turnover, as an experienced leaver is replaced by a low-skill junior.
Attrition and tenure	Cumulative number of departures and the team’s mean tenure (ticks since each current member was hired).

7 Implementation and methods

7.1 Software and code organisation

The model is implemented in Python on top of Mesa 3.x, the standard agent-based-modelling framework; figures are rendered with Matplotlib, the network animation with Plotly, and the interactive front-end with Streamlit. The code separates the simulation itself from everything that drives or observes it (Table 4): one self-contained model module, a set of command-line experiment harnesses, analysis and plotting scripts, and a test suite. The whole model is about 560 lines; no run depends on any state outside its own `TeamModel` instance.

Table 4: Code organisation: the simulation is isolated from the harnesses that drive it and the scripts that analyse its output.

File	Role
<code>model.py</code>	Core ABM: <code>TeamModel</code> (Mesa <code>Model</code>), <code>Dev</code> (Mesa <code>Agent</code>), and <code>Task</code> (a passive <code>dataclass</code>).
<code>experiment.py</code>	Generic <code>remote_share</code> sweep harness; each run yields one summary-metrics row.
<code>run_experiments.py</code>	The three single-sprint sweeps of Part A (baseline, $\times$ skill, $\times$ blocker rate).
<code>run_extended.py</code> , <code>analyze_-</code> <code>extended.py</code>	The four year-long sweeps of Part B.
<code>sensitivity_-</code> <code>remote.py</code>	Parallel one-parameter sensitivity study; raw per-run output.
<code>calibrate_-</code> <code>learning.py</code>	Fits <code>learning_rate</code> to the literature ramp-up curve.
<code>plot_*.py</code> , <code>analyze*.py</code>	Aggregation (means, CIs) and figure rendering.
<code>app_streamlit.py</code>	Interactive front-end over the same model.
<code>tests/</code>	<code>pytest</code> suite (87 tests across 14 files).

7.2 Simulation engine

A run advances tick by tick. On each tick the model re-rolls daily office/home locations at day boundaries, closes the sprint and resolves turnover at sprint boundaries, and then activates the agents. Activation is deliberately *two-phase*: in the first pass every developer that is not currently seeking help takes its action (work, idle pick-up, solo attempt, help-session tick), and

in the second pass the help-seekers each choose a helper from the now-stable pool of available colleagues. Without this split, whether a recruited helper had already worked that tick would depend on the random order in which agents were visited, making runs irreproducible.

Reproducibility is treated as a contract. A run is driven by a single seeded random stream (Mesa’s `Model(seed=...)`), and every stochastic step consumes a fixed number of draws in a fixed order regardless of outcome: `turnover`, for instance, performs exactly one draw per developer in list order whether or not anyone leaves. A given parameter set and seed therefore reproduce a run exactly, which the test suite checks directly. Learning, by contrast, is fully deterministic: the bounded geometric update toward `skill_cap` uses no randomness, so its trajectory is identical across seeds. Each tick a Mesa `DataCollector` records about sixteen macroscopic reporters (velocity, mean wait, the per-state head-counts, team skill, attrition, mean tenure, and so on), which are the series plotted in the interface and aggregated by the analysis scripts. Parameters are validated at construction, so out-of-range values fail fast rather than propagating into a run.

7.3 Experimental method and testing

Every experiment follows the same Monte-Carlo pattern: each parameter cell is run for many replications under distinct, deterministic seeds (a reproducible function of the cell and replication index), and the *raw per-run* results are written to CSV so that the full run-to-run distribution, not just its mean, survives for later analysis. Aggregation and figure rendering live in separate scripts, so the same CSVs can be re-analysed without re-running the model. The single-sprint and year-long sweeps run sequentially; the sensitivity study, far larger at roughly 3,000 runs, is parallelised across worker processes with Python’s `multiprocessing`, each worker executing one independent simulation. Correctness is guarded by a `pytest` suite of 87 tests across fourteen files, covering the state machine and blocker handling, the synchronous/asynchronous help channels, the learning and turnover mechanics, multi-sprint horizons, the two-phase activation, data collection, parameter validation, and edge cases, with a dedicated group asserting bit-for-bit determinism under a fixed seed.

8 Discussion and conclusions

The model does not deliver a verdict that remote work is simply good or bad for output. Its central result is that the cost of remote work is *contingent on the state of the team*. Pulling the threads together:

- **The remote penalty is a tax on the help channel.** Over a single sprint it scales with how heavily that channel is used: small (-3.6%) for a high-skill team, large (-13.2%) for a team that blocks frequently. A team that rarely needs help barely notices remote work; a junior, blocker-heavy team pays dearly.
- **Learning attenuates the penalty rather than compounding it.** Over a year the office advantage is only $\sim 2.5\%$ and is front-loaded, because as both teams climb the learning curve they use the penalised help channel less. This emergent reversal of the naive prior is a direct argument for the agent-based method over a static estimate.
- **The decision-relevant effect is on risk, not on the mean.** Going fully remote leaves expected yearly output almost unchanged but makes it about three times less predictable. A recommendation based on averages alone would miss the variable that a risk-aware manager cares about.
- **Retention is the open question.** Whether remote work helps or harms retention is genuinely unresolved in the literature; the model carries both readings as named scenar-

ios rather than choosing one, and over multiple years this channel could dominate the throughput effect studied here.

A practical reading is therefore conditional rather than universal: remote work is cheap for a mature, stable, high-skill team and costly for a junior-heavy, high-churn team that is still onboarding, where the cost front-loads exactly when the team can least afford it.

Limitations and next steps. No published source measures the model’s exact constructs, so the parameters are reconstructions and the results should be read as the shape of a response, not as calibrated forecasts. The model uses a single, undifferentiated `remote_attrition_coef` and so cannot capture the seniority-conditioning the literature shows; the learning-channel coupling is muted at the default low blocker rate and would become material only in a blocker-heavy or junior-heavy regime. The clearest next step is a full sweep of `remote_attrition_coef` over $[-1, 1]$, which would couple retention back into the skill channel over a multi-year horizon and plausibly dominate the 2.5% throughput effect measured here.

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